# Project Report: Blockchain-Based Voting System

Team BBVS

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# 1 Abstract

This report presents the development and implementation of a Blockchain-Based Voting System (BBVS), a secure and transparent platform for conducting online elections. Leveraging blockchain technology, specifically Ethereum, alongside a modern web stack, the system ensures vote integrity, transparency, and accessibility. The project integrates a React and TypeScript frontend, a Node.js and TypeScript backend, a MySQL database, and Ethereum smart contracts deployed via Truffle. This report outlines the project objectives, system architecture, implementation details, challenges faced, results achieved, and future scope.

# 2 Introduction

The Blockchain-Based Voting System (BBVS) addresses the need for secure, transparent, and verifiable online voting. Traditional voting systems often face challenges such as fraud, lack of transparency, and accessibility issues. By utilizing blockchain technology, BBVS ensures that votes are immutable, verifiable, and securely recorded. The system is designed for ease of use, with an intuitive frontend interface and robust backend infrastructure.

## 2.1 Objectives

- Develop a secure voting platform using Ethereum blockchain for vote storage.
- Ensure transparency and verifiability of votes through blockchain immutability.
- Provide an accessible and user-friendly interface for voters and administrators.
- Implement a scalable backend with secure authentication and data management.
- Address common voting system vulnerabilities, such as vote tampering and unauthorized access.

# 3 System Architecture

The BBVS is structured as a full-stack web application with blockchain integration. The architecture comprises four main components:

## 3.1 Frontend

- **Technology**: React, TypeScript, Bootstrap
- **Functionality**: Provides user interfaces for voter registration, poll creation, vote casting, and result viewing. The landing page allows navigation to admin and voter functionalities.
- **Key Features**: Responsive design, form validation, and real-time vote updates.

## 3.2 Backend

- **Technology**: Node.js, TypeScript, TypeORM

- **Functionality**: Handles API requests, user authentication (JWT-based), and database interactions. It interfaces with the blockchain for vote storage.
- **Key Features**: RESTful APIs, secure token-based authentication, and database migrations.

### 3.3 Database

- **Technology**: MySQL
- **Structure**: Stores user data (admin and voters), poll details, and candidate information. Tables include `user`, `poll`, and `candidate`.
- **Configuration**: Secured with a dedicated `bbvs` user and restricted privileges.

### 3.4 Blockchain

- **Technology**: Ethereum, Truffle, Ganache (for local testing)
- **Functionality**: Smart contracts store votes immutably, ensuring transparency and verifiability. Truffle manages contract deployment and migrations.
- **Key Features**: Decentralized vote storage, public verification of results.

## 4 Implementation Details

The BBVS was developed following a structured process, with distinct phases for setup, development, and testing.

### 4.1 Setup

- **Environment**: Node.js (v16), MySQL (v8.0.42), and Truffle were installed. Ganache CLI provided a local Ethereum blockchain.
- **Database**: The `bbvs` database was created with a secure user (`bbvs@localhost`). TypeORM migrations initialized the schema.
- **Blockchain**: Smart contracts were written in Solidity, compiled, and deployed using Truffle.

### 4.2 Development

- **Frontend**: React components (e.g., `AdminLogin.tsx`, `LandingPage.tsx`) were built with TypeScript for type safety. Bootstrap ensured a responsive UI.
- **Backend**: Node.js APIs were developed for user management, poll creation, and vote submission. TypeORM handled database operations.
- **Blockchain Integration**: Smart contracts were integrated with the backend using Web3.js, enabling vote recording on the Ethereum blockchain.

### 4.3 Testing

- **Unit Testing**: Frontend components were tested for rendering and form validation. Backend APIs were tested for correct responses.
- **Integration Testing**: Ensured seamless interaction between frontend, backend, database, and blockchain.
- **Security Testing**: Verified JWT authentication and blockchain vote integrity.

## 5 Challenges Faced

Several challenges were encountered during development:

- **MySQL Syntax Errors**: Initial database setup had privilege errors (e.g., `GRANT ALL ON *.*`). Resolved by restricting privileges to the `bbvs` database.
- **Truffle Migration Issues**: Errors with `uWS` dependencies were fixed by reinstalling compatible versions.
- **npm Vulnerabilities**: Deprecated packages (e.g., `uuid`, `svgo`) and vulnerabilities were addressed using `npm audit fix` and manual updates.
- **Blockchain Gas Costs**: High gas costs for contract deployment were mitigated by optimizing Solidity code.

## 6 Results

The BBVS was successfully implemented and tested locally. Key outcomes include:

- **Secure Voting**: Votes were immutably recorded on the Ethereum blockchain, ensuring no tampering.
- **Transparency**: Voters and administrators could verify results via blockchain explorer or the frontend interface.
- **Usability**: The intuitive UI allowed easy navigation for poll creation, voting, and result viewing.
- **Scalability**: The backend handled multiple concurrent users during testing.
- **Admin Features**: Admins could manage users and polls efficiently (e.g., using credentials `john@gmail.com`).

## 7 Future Scope

The BBVS has potential for further enhancements:

- **Scalability**: Deploy on a public Ethereum network or a layer-2 solution (e.g., Polygon) to reduce gas costs.
- **Accessibility**: Add multi-language support and accessibility features for visually impaired users.

- **Security**: Implement two-factor authentication and advanced encryption for user data.
- **Mobile App**: Develop a mobile application for broader voter access.
- **Analytics**: Integrate dashboards for real-time voting analytics and trends.

## 8 Conclusion

The Blockchain-Based Voting System demonstrates the potential of blockchain technology in creating secure, transparent, and accessible voting platforms. By combining modern web technologies with Ethereum smart contracts, BBVS addresses critical challenges in traditional voting systems. Despite initial setup and dependency issues, the system was successfully deployed and tested, offering a robust solution for online elections. Future enhancements will further improve its scalability and usability.

## 9 References


- Ethereum Documentation: `https://ethereum.org`
- Truffle Suite: `https://trufflesuite.com`
- React Documentation: `https://reactjs.org`
- Node.js Documentation: `https://nodejs.org`
- MySQL Documentation: `https://dev.mysql.com/doc`